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This is a Author's accepted manuscript (AAM) version of a publication  
published by IEEE

in IEEE Journal of Biomedical and Health Informatics

**DOI:** 10.1109/JBHI.2024.3366341

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### **Please cite the publication as follows:**

I. Ahmad, M. Zhu, G. Li, D. Javeed, P. Kumar and S. Chen, "A Secure and Interpretable AI for Smart Healthcare System: A Case Study on Epilepsy Diagnosis Using EEG Signals," in IEEE Journal of Biomedical and Health Informatics, vol. 28, no. 6, pp. 3236-3247, June 2024, doi: 10.1109/JBHI.2024.3366341

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# A Secure and Interpretable AI for Smart Healthcare System: A Case Study on Epilepsy Diagnosis using EEG Signals

Ijaz Ahmad, Mingxing Zhu, Guanglin Li, Danish Javeed, Prabhat Kumar, and Shixiong Chen\*

## Abstract

The efficient patient-independent and interpretable framework for electroencephalogram (EEG) epileptic seizure detection (ESD) has informative challenges due to the complex pattern of EEG nature. Automated detection of ES is crucial, and Explainable Artificial Intelligence (XAI) is urgently needed to justify algorithmic predictions in clinical settings. Therefore, this study implements an XAI-based computer-aided ES detection system (XAI-CAESDs), comprising three major modules including of feature engineering module, a seizure detection module, and an explainable decision-making process module in a smart healthcare system. To ensure the privacy and security of biomedical EEG data, the blockchain is employed. Initially, the Butterworth filter eliminates various artifacts, and the Dual-Tree Complex Wavelet Transform (DTCWT) decomposes EEG signals, extracting real and imaginary eigenvalue features using frequency domain (FD), time domain (TD), and Fractal Dimension (FD) of linear and non-linear features. The best features are selected by using Correlation Coefficients (CC) and Distance Correlation (DC). The selected features are fed into the Stacking Ensemble Classifiers (SEC) for EEG ES detection. Further, the Shapley Additive Explanations (SHAP) method of XAI is implemented to facilitate the interpretation of predictions made by the proposed approach, enabling medical experts to make accurate and understandable decisions. The proposed ensemble-based stacking classifiers in XAI-CAESDs have demonstrated 2% best average accuracy, Recall, specificity, and F1-score using the University of California, Irvine, Bonn University, and Boston Children's Hospital-MIT EEG data sets. The proposed framework enhances decision-making and the diagnosis process using biomedical EEG signals and ensures data security in smart healthcare systems.

## Index Terms

Biomedical data, explainable artificial intelligence, smart healthcare, data security, epileptic seizures

## I. INTRODUCTION

Epilepsy is a neurologist's chronic disorder disease that occurs due to a sudden abnormality in the brain's electrical activity [1]. Epilepsy is classified as the fourth most prevalent brain disorder disease, affecting more than 50 million people worldwide [1]. An electroencephalogram (EEG) is a highly effective tool for examining the brain's electrical activity [2]. It is accomplished via an electrode position, either outside (extracranial recording) or inside the skull (intracranial recording). Traditionally, medical experts visualize the EEG signals using their domain expertise. However, manual visualization is susceptible to human error, time-consuming, labor-intensive, and often inadequate for accurate information. Therefore, the development of automatic EEG epileptic seizure detection (ESD) has significant importance for researchers in the field of clinical neurology [3]. In contrast, accurate ES detection serves as a crucial element for automated closed-loop treatment, in which interventions such as cooling, drug infusion, and electrical stimulation can be administered to patients upon seizure diagnosis [4]. Traditional automatic EEG epileptic seizure detection can be divided into three main components: feature extraction, feature selection, and classification. Various features, such as the time domain feature [4], frequency domain feature [5] of linear while non-linear features like Fractal Dimension Features [6], are extracted from EEG signals. These features can be obtained directly from EEG signals or decomposed EEG signals using techniques like Discrete Wavelet Transform (DWT) [7] and Continuous Wavelet Transform (CWT) [8]. After feature extraction, efficient feature selection methods, such as correlation coefficient [9], and mutual information-based selection methods [10] are utilized to select the optimal features. Once the best features are determined, they are passed through various machine learning (ML) models, including Decision Tree (DT) algorithms, Random

This work was supported in part by the National Key R&D Program of China (2022YFE0197500), National Natural Science Foundation of China (#81927804, #62101538), Science and Technology Planning Project of Shenzhen (#JSGG20210713091808027, #JSGG20211029095801002), China Postdoctoral Science Foundation (2022M710968). (The first two authors contributed equally to this work.) (Corresponding author: Shixiong Chen)

I. Ahmad, M. Zhu, G. Li are with the CAS Key Laboratory of Human-Machine Intelligence Synergy Systems, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong 518055, China. (Email: ijaz@siat.ac.cn, bestarpc@126.com, gl.li@siat.ac.cn)

I. Ahmad is with the Shenzhen College of Advanced Technology, University of Chinese Academy of Sciences, Shenzhen, Guangdong 518055, China.

I. Ahmad, M. Zhu, G. Li are also with the Guangdong-Hong Kong-Macao Joint Laboratory of Human-Machine Intelligence-Synergy Systems, Shenzhen, Guangdong 518055, China.

M. Zhu is also with the School of Electronics and Information Engineering, Harbin Institute of Technology, Shenzhen, Guangdong 518055, China.

D. Javeed is with the Software College, Northeastern University, Shenyang 110169, China. (Email: 2027016@stu.neu.edu.cn).

P. Kumar is with the Department of Software Engineering, LUT School of Engineering Science, LUT University, 53850 Lappeenranta, Finland (Email: prabhat.kumar@lut.fi).

S. Chen is with the School of Medicine, The Chinese University of Hong Kong, Shenzhen, Guangdong 518172, China. (Email: chenshixiong@cuhk.edu.cn)

Forest (RF) classifiers, Gradient Boost classifiers (GB), Logistic regression (LR) Regression, and stacking ensemble classifiers (SEC), to achieve accurate and computationally efficient ESD [11], [12].

However, the existing EEG ESD frameworks often lack real-time performance, especially in terms of specificity and sensitivity, limiting their applicability in clinical practice and facing several informative challenges. The primary challenge lies in the decomposition of EEG signals, where transformation coefficients have unpredictable changes with slight shifts in the EEG signal, often failing to represent the changes accurately [13]. After decomposition, the second challenge is extracting meaningful and informative features that can effectively distinguish between different EEG states during seizures, while in feature extraction, the third challenge involves selecting the best features. The process necessitates considering multiple scales and coefficients to capture seizure patterns, as well as computing the complexity and variability of these patterns across different seizure types. Feature selection requires careful consideration of factors like noise mitigation, redundant features, and the limited availability of labeled data. The fourth challenge involves reducing the false positive rate, which can arise from the same pattern of oscillatory characteristics between ictal and interictal EEG signals, while The fifth challenge pertains to the lack of studies demonstrating the explainability and interoperability of the model output for EEG ES detection while creating patients' independent frameworks is also a challenging task because each patient has a different pattern of seizures.

To overcome the aforementioned challenges, this study introduces a novel patient-independent pipeline for automatic EEG ES detection using biomedical EEG signals. The contributions of the proposed study are given below:

- Construction of EEG signal decomposition, segmentation, and feature extraction using the dual-tree complex wavelet transformation (DTCWT) has been applied at various levels for feature extraction, and statistical, clinical, and nonlinear features are obtained from the complex-valued feature vector.
- Construction of improved distance and coefficient correlation, which selects the optimal linear and non-linear feature from the real and imaginary decomposition levels and overcomes the problems of overestimation of feature significance, redundancy, and irrelevant features.
- Stacking Ensemble-Based Classifiers is developed. The base model uses a combination of bagging random forest (RF), decision tree (DT), and Extreme Gradient Boosting (XGB) to find complex patterns in the feature data. The meta-classifier logistic regression (LR) acts as a meta-model, providing a comprehensive solution for EEG-based seizure detection. The blockchain is employed to ensure the security and privacy of biomedical data.
- Interpretability and Explainability employ SHAP to offer insights into model decisions, emphasizing the importance of features and bolstering trust in the model's outcomes.

The proposed framework offers various advantages. As tested on the UCI, CHB-MIT, and Bonn EEG datasets, it optimizes accuracy by selecting efficient features while reducing over-fitting. The framework of novel feature selection algorithms and stacking tree-based ensemble classifiers improved the classification accuracy in EEG epileptic seizure detection, while explainable AI (XAI) increased the model's transparency, offering clinicians insights into its decisions and ensuring better epilepsy diagnosis and decision-making.

This paper is organized as follows: Sections I and II present the introduction and related works, respectively. Sections III introduce the novel pipeline framework for EEG ES detection. The discussion is in sections IV, while the results are described in sections V respectively. Finally, the conclusion of the study is in Section VI.

## II. RELATED WORK

This section explains recent literature that focuses on decomposition methods of the EEG signals, linear and non-linear features, and feature selection techniques with machine learning (ML) models in EEG epileptic seizure detection (ESD) [7], [8].

Zhang et al. [14] implemented several wavelet-based features to classify EEG signals. They pass the EEG signals into various preprocessing steps. Subsequently, features such as HE, HFD, and Shannon entropy (SE) were extracted and demonstrated effectiveness with ML models like RF and SVM. Nasir et al. [15] employed the Pearson correlation coefficient (PCC) for feature selection and achieved 93.10% accuracy. Mursalin et al [16] further refined this approach, optimizing selection from entropy, frequency, time features, and wavelet decomposition and pairing it with an RF model for effective classification. Moreover, Staking ensemble methods, a common model in ensemble learning, have found widespread application in EEG epileptic seizure detection. Sun et al. [17] analyzed the experimental results of three prominent ensemble techniques: boosting, bagging, and random ensembles. Their findings emphasized that the effectiveness of the ensemble approach is contingent upon the choice of base classifiers, particularly the configurations and parameters employed for each classifier. Dehuri et al. [18] introduced the ensemble of Radial Basis Function Neural Networks (RBFNs) as a method for identifying epileptic seizures. This approach was grounded in the bagging methodology and employed Differential Evolution (DE) RBFNs as the underlying base classifier. He et al. [19] put forth a Signal Strength-Based Combining (SSC) method aimed at classifying EEG ES detection. Their results demonstrated that the proposed SSC method holds its own against existing classifiers and offers competitive performance. Wang et al. [20] introduced a model tailored for biomedical ECG and EEG time series, representing them as histograms of code words. Notably, the results obtained with this model exhibited resilience to parameter variations and robustness in the presence of noise. Most of the mentioned studies applied DWT which is an effective tool for capturing and

localizing transient changes in EEG recordings [7]. However, one crucial drawback that has been previously highlighted when using DWT is their susceptibility to shift variance. Even small shifts in the EEG signal can significantly affect the resulting wavelet coefficients [7], [13]. It is well established that the dual-tree complex wavelet transform (DTCWT) technique, which employs specially designed filter banks [21] for wavelet-domain processing, effectively mitigates the above problem. Notably, there is limited reported research on processing epilepsy signals using DTCWT [22]. This work demonstrates the extraction of time-frequency domain features from normal and seizure signals using the DTCWT method [22]. Moreover, the above studies inspire researchers to employ linear features from statistical, frequency domains, and Fractal Dimensions (FDs) as non-linear features, along with ensemble ML models to efficiently detect EEG-based epileptic seizure.

Table I: The detailed description of the Bonn EEG dataset.

| Subject           | Five healthy subject |                     | Five epileptic subject    |                           |                          |
|-------------------|----------------------|---------------------|---------------------------|---------------------------|--------------------------|
| Experimental type | Set A                | Set B               | Set C                     | Set D                     | Set E                    |
| Patient state     | Eyes Open (Normal)   | Eyes Close (Normal) | Seizure-free (Interictal) | Seizure-free (Interictal) | Seizure activity (Ictal) |
| Number of files   | 100                  | 100                 | 100                       | 100                       | 100                      |
| Electrode Type    | Surface              | Surface             | Surface                   | Intracranial              | Surface                  |
| Sample length     | 4097                 | 4097                | 4097                      | 4097                      | 4097                     |

Table II: The summary of the UCI epileptic seizure recognition data set.

| EEG Data set                   | The public UCI               |
|--------------------------------|------------------------------|
| <b>Patients</b>                | 5 (4 seizure, 1 non-seizure) |
| <b>Type</b>                    | Scalp EEG                    |
| <b>Total sample</b>            | 11500                        |
| <b>EEG recording</b>           | 23.6 sec                     |
| <b>Sampling Rate/frequency</b> | 173.6 Hz                     |
| <b>Electrode Positions</b>     | 10-20 method                 |

Table III: The detailed information of the CHB-MIT EEG dataset. Gender: Male (M) and Female (F). No. of seizure: The number of seizures.

| Patient ID | Gender | Age  | No. of seizure | Hours of sEEG |
|------------|--------|------|----------------|---------------|
| 1          | F      | 11   | 7              | 41            |
| 2          | M      | 11   | 3              | 35            |
| 3          | F      | 14   | 7              | 38            |
| 4          | M      | 22   | 4              | 156           |
| 5          | F      | 7    | 5              | 39            |
| 6          | F      | 1.5  | 10             | 67            |
| 7          | F      | 14.5 | 3              | 67            |
| 8          | M      | 3.5  | 5              | 20            |
| 9          | F      | 10   | 4              | 68            |
| 10         | M      | 3    | 7              | 50            |

### III. CASE STUDY: EEG-BASED EPILEPTIC SEIZURE DETECTION (ESD)

In this section, we describe the proposed novel framework, which includes three main modules: feature engineering modules, seizure detection, and explainable decision-making process for the proposed model, as illustrated in Fig. 1.

#### A. Feature engineering module

In the feature engineering module, we discuss the process from EEG data collection to feature selection using different methods.

1) *Bonn EEG dataset (DS1) [23]*: The Bonn EEG dataset used for epileptic seizure classification includes five sets labeled A to E. Each set contains 100 single-channel EEG recordings of 23.6 seconds, using a 173.6 Hz sampling frequency and 12-bit resolution. Sets A and B, from healthy individuals, represent eyes-open and eyes-closed conditions, respectively. Sets C, D, and E are from epileptic patients and have inter-ictal, pre-ictal, and ictal EEG states, respectively. Sets A and B are seizure-free, while C, D, and E are transitional or interictal states, and E contains seizure or ictal data. Each set has 4097 samples, as shown in Table I.

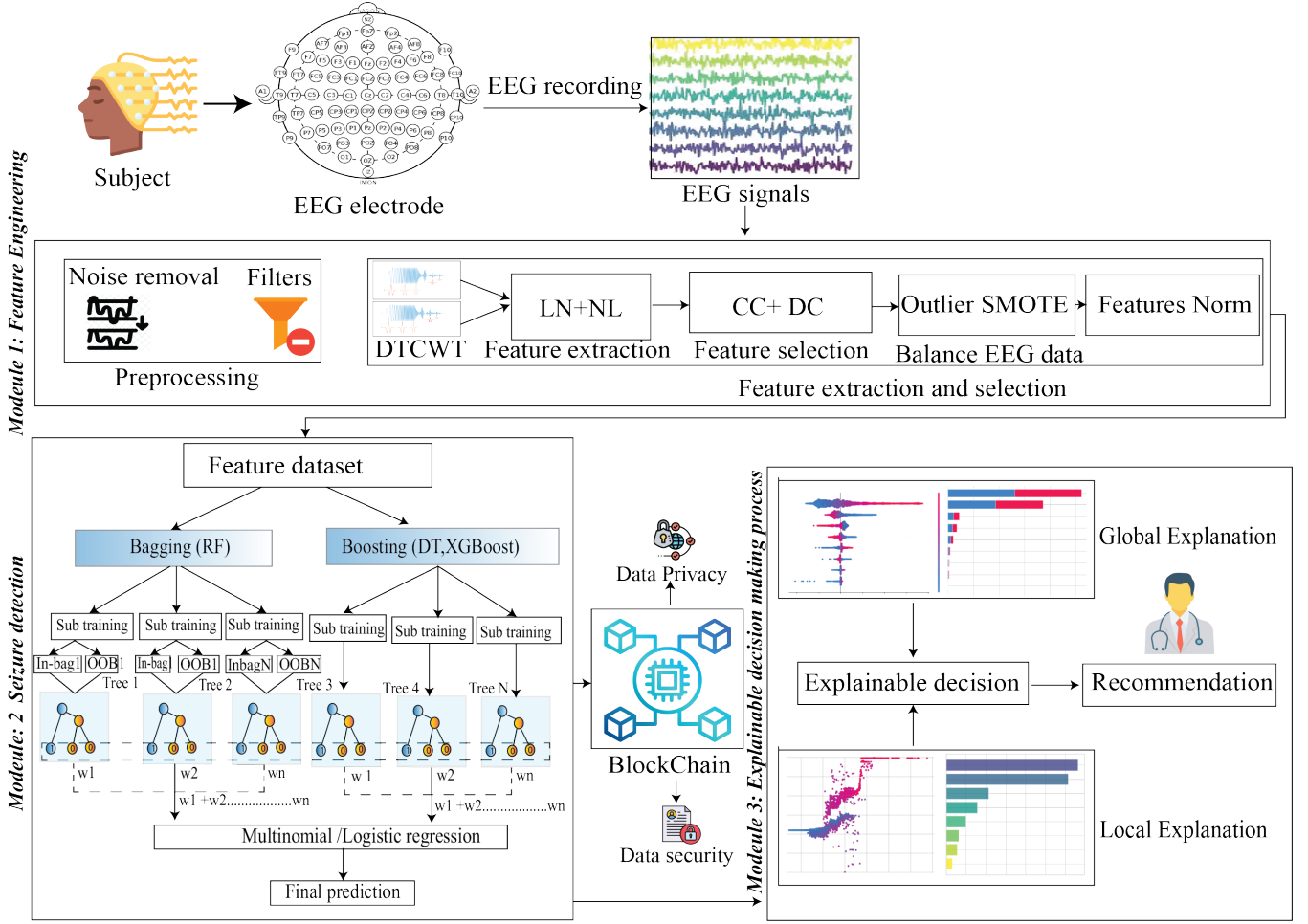


Figure 1: Schematic illustration of the overall approach pipeline. The framework consists of feature engineering module, seizure detection module, and an explainable decision-making process module. The Blockchain is implemented for the security and privacy of the proposed model outcomes.

2) *UCI EEG dataset (DS2)* [24]: The proposed UCI EEG dataset for epileptic seizures is a widely recognized public dataset. It's organized into five directories, each file has 100 sub-files, and every sub-file record has brainwave EEG data for 23.6 seconds. The dataset includes a total of 11,500 instances, a total of 23 segments per 500 files, and 178 features per segment. The data is categorized into five binary classes of different EEG states, labeled as  $y=1,2$  and  $y=1, 2, 3, 4, 5$ . Each EEG state consists of the 4097 sample length. The sampling frequency is 173.6 Hz. Each file contains 2300 data points. The seizure-free sample has 11500 samples, while the seizure-free sample has 2300 samples. The data is balanced using the outlier SMOTE (Synthetic Minority Oversampling Technique). Set A shows the seizure activity, while from SET B to SET E are normal activities of the brain.

3) *CHB-MIT EEG dataset (DS3)* [25]: The CHB-MIT dataset comes from a children's hospital in Boston and includes data from 22 patients aged between 1.5 and 22 years. Each patient's data consists of 9–42 continuous EDF (European Data Format) files and each file includes a complete one-hour recording. The sampling frequency was 256 Hz. In the proposed study, the first 10 subjects are used for the experiment, which has a 4-hour interictal duration between seizures while the maximum number of seizures is greater than 3. All the seizures and normal of the patients are categorized into set (A) seizure and set (B) normal. The detailed explanation of all the subjects is explained in the given literature [25].

### B. EEG data preprocessing

All the EEG datasets contain noise and artifacts resulting from factors such as eye blinking, EMG, and changes in limb position, which contribute to undesirable high frequencies. To address these artifacts and noise, we apply the Butterworth filter, as detailed below;

$$X(t)_{k-l} = BWF(y(t), k, l), k = 0, l = 100 \quad (1)$$

Where  $k$  and  $l$  show the range of frequency, while  $y(t)$  shows the signals to be filtered. Besides, to address sample imbalances of the UCI and CHB-MIT EEG datasets. A Butterworth filter is utilized and the frequency ranges from 0.1-30Hz, the filter

Table IV: Notation Table

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| $\phi_h(t)$          | Hilbert transform of the wavelet                                |
| $\psi_g(t)$          | Imaginary-based decomposition                                   |
| $g_0[n]$             | Distance between data points                                    |
| $X_i$                | Individual data points                                          |
| $f(x)$               | Nonlinear function                                              |
| $(A, D)$             | The joint characteristic function of the pair of random vectors |
| $L$                  | Length                                                          |
| $\ y\ $              | The Euclidean norm                                              |
| $\phi_A(x)\phi_D(y)$ | Mutual independent                                              |
| $V^2(A, D)$          | Squared distance covariance                                     |
| $n$                  | Total number of points                                          |
| $g_0, g_1$           | Low pass filter and high pass filter                            |

settings include a cutoff frequency of 0.1, a filter order of 3, and a Nyquist frequency of 0.5. To address sample imbalances in the UCI, CHB-MIT, and Bonn EEG datasets, the Outlier Synthetic Minority Oversampling Technique (SMOTE) is utilized.

### C. Signals decomposition and feature extraction

Discrete Wavelet Transformation (DWT) is extensively applied across various domains such as signal and image compression, feature extraction, noise reduction, and channel coding [26]–[28]. Despite its efficient computational approach and accurate representation, DWT is limited by several limitations compared to the Dual-Tree Complex Wavelet Transform (DTCWT). The DWT is sensitive to shifts in the time domain of the input EEG signal, meaning that transformation coefficients can change unpredictably with slight shifts in the EEG signal. Due to this disadvantage, DWT coefficients often fail to accurately represent changes in the EEG signal [29]. In contrast, DTCWT is less sensitive to such shifts, offering improved time-shift responsiveness. The standard DWT lacks effectiveness in identifying diagonal features in signals. In the proposed study, DTCWT, exhibits better direction with its 10 wavelets including five real and imaginary levels in two dimensions, while the limitation of DWT is the lack of phase information. Since most natural EEG signals are real-valued, complex-valued filtering is necessary to utilize local phase information [30]. The DTCWT, employing complex functions rather than real-valued wavelet functions, allows for the separate analysis of amplitude and phase information. DTCWT operates using analytic filters for wavelet analysis and is structurally more complex than standard DWT, consisting of two parallel DWTs called the imaginary and real decomposition levels of the wavelets [30]. The DTCWT employs two filters for the main wavelet function and the scaling function, which include  $A_0(n)$  and  $A_1(n)$  is the low-pass and high-pass filters, and  $h_0(n)$  and  $h_1(n)$  as another set of low-pass and high-pass filters [31]. The wavelet and scaling functions in DTCWT are calculated accordingly. If is nearly analytic (supported primarily on one side of the frequency axis), the resulting transformation can exhibit characteristics like lack of aliasing, directionality, and shift in variance, similar to the Fourier transform [31]. For approximately analytic, one wavelet is always the same as another wavelet of the Hilbert transform.

$$\psi_h(a) = \sqrt{2} \sum_n h_1(n) \phi_h(2a - n) \quad (2)$$

$$\phi_h(a) = \sqrt{2} \sum_n h_0(n) \phi_h(2a - n) \quad (3)$$

This is designed to provide the Hilbert transformation as per [31]. Condition for the Hilbert Transform:

$$\psi_g(t) \approx H\{\psi_h(t)\}. \quad (4)$$

To fulfill the condition in equation (3), the low-pass filter analysis in the real and imaginary trees should have approximately half-sample distance:

$$g_0[n] \approx h_n[n - 0.5]. \quad (5)$$

We utilized the q-shift filter as used in the literature [30]. Additional information about the q-shift filtering process can be found in [31]. For the feature extraction, the Bonn EEG dataset contains (100 Segments×4096 samples) from A set to set E, while the UCI EEG dataset represents 2300 samples for the seizure, while 11500 samples for the non-seizure, later Outlier

SMOTE balances the seizure data, and for the CHB-MIT EEG dataset 8976 segments of 23 channel each channel is 256 Hz of one-second data by three-level decomposition of both real and imaginary components are implemented, ranging from D1 to D3. The statistical time-domain features such as variance (Var), standard deviation (SD), skewness (Skew), and mean. Statistical frequency-domain features such as Approximation entropy (AE), Permutation Entropy (PE), and spectral entropy (SE), while non-linear features include fractal dimension (FD), and Higuchi's Fractal Dimension (HFD) are extracted. More explanation of the linear and nonlinear features used in the proposed study can be found in [5]. Moreover, Table IV represents the mathematical notation and description of the proposed study.

$$\text{Mean} = \mu = \frac{1}{N} \sum_{i=1}^N X_i \quad (6)$$

where  $\mu$  is the mean,  $N$  shows total number of signals data, and  $X_i$  are the individual data points.

$$\text{SD} = \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2} \quad (7)$$

where  $\sigma$  is the standard deviation,  $X_i$  are the EEG data, and  $\mu$  is the mean.

$$\text{Var} = \sigma^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2 \quad (8)$$

$$\text{Skew} = \frac{\sum_{i=1}^N (X_i - \mu)^3}{(N - 1) \cdot \sigma^3} \quad (9)$$

$$\text{FD} = \frac{\log(n)}{\log(d/L) + \log(n)} \quad (10)$$

where  $d$  is the maximum Euclidean distance, and  $L$  is the total length of the waveform.

$$\text{SE} = - \sum_{i=1}^N p_i \log_2(p_i) \quad (11)$$

where  $p_i$  represents the normalized power spectral density for each frequency bin  $i$ .

$$\text{PE} = - \sum c \log_2(c) \quad (12)$$

where  $c$  represents the normalized frequency of each permutation pattern observed in the time series.

$$\text{HFD} = \frac{\log_{10}(d(e))}{\log_{10}(1/e)} \quad (13)$$

where  $d(e)$  shows the curve distance at different scales and  $k$  is the scale size.

#### D. Feature selection methods

To determine the strength and direction of the association between variables, the Pearson correlation coefficient is utilized. The statistical measure ranges from -1 for a perfect negative correlation to 1 for a strong positive correlation. Based on our findings, any features correlating to more than 0.8 are removed. To establish the statistical significance of relationships between features and data types, a threshold of  $pvalue < 0.05$  is set. Subsequently, after completing all eliminations, the 60 best eigenvalue linear features encompassing both time and frequency domain features are selected from the features of the Bonn EEG experimental type (AB, CD, E). The UCI-EEG case (A, B) from the CHB-MIT EEG dataset are selected.

After the correlation coefficient (CC), the distance correlation (DC), measures the distance between the feature values, or characteristic functions to evaluate the dependency between sets of random variables. The method plays a crucial role in delineating the independence of the variables. let's consider a positive integer  $p$  and a random vector  $A = (A_1, \dots, A_p)$  in  $\mathbb{R}^p$ . For a vector  $x = (x_1, \dots, x_p)$  in  $\mathbb{R}^p$ , the Euclidean norm  $\|x\| = \sqrt{x_1^2 + \dots + x_p^2}$  represents the standard Euclidean norm on  $\mathbb{R}^p$ . The standard inner product between  $x$  and  $A$  is denoted as  $\langle x, A \rangle = x_1 A_1 + \dots + x_p A_p$ . Similarly, consider another positive integer  $q$ , a vector  $y$  in  $\mathbb{R}^q$ , and a random vector  $D$  in  $\mathbb{R}^q$ , associated with  $A$ . The Euclidean norm  $\|y\|$  and the inner product  $\langle y, D \rangle$  on  $\mathbb{R}^q$  are defined in a similar manner. The joint characteristic function of the pair of random vectors  $(A, D)$  is then defined as:

$$\phi_{A,D}(x, y) = \mathbb{E}[\exp(i\langle x, A \rangle + i\langle y, D \rangle)] \quad (14)$$

The marginal characteristic functions of  $A$  and  $D$  are:

$$\phi_A(x) = \phi_{A,D}(x, 0) = \mathbb{E}[\exp(i\langle x, A \rangle)], \quad (15)$$

$$\phi_D(y) = \phi_{A,D}(0, y) = \mathbb{E}[\exp(i\langle y, D \rangle)] \quad (16)$$

respectively.  $A$  and  $D$  are mutually independent if and only if  $\phi_{A,D}(x, y) = \phi_A(x)\phi_D(y)$  for all  $x \in \mathbb{R}^p$  and  $y \in \mathbb{R}^q$ . The distance covariance between  $A$  and  $D$  is a non-negative number  $V(A, D)$ , defined as:

$$V^2(A, D) = \frac{1}{c_p c_q} \int_{\mathbb{R}^q} \int_{\mathbb{R}^p} |\phi_{A,D}(x, y) - \phi_A(x)\phi_D(y)|^2 \|x\|^{p+1} \|y\|^{q+1} dx dy \quad (17)$$

where  $V^2(A, D)$  represents the squared distance covariance between the random vectors  $A$  and  $D$ , while  $\phi_{A,D}(x, y)$  is the joint characteristic function of  $A$  and  $D$ , where  $\phi_A(x)$  and  $\phi_D(y)$  are the marginal characteristic functions of  $A$  and  $D$ , respectively.  $\|x\|$  and  $\|y\|$  denote the Euclidean norms of vectors  $x$  and  $y$ . The integration is carried out over the entire space of  $\mathbb{R}^p$  and  $\mathbb{R}^q$ ,  $c_p$  and  $c_q$  are constants depending on the dimensions  $p$  and  $q$  of the vectors  $A$  and  $D$ .

---

**Algorithm 1** Distance correlation coefficient for feature selection method

---

```

1: function IMPROVE_DISTCORR(Data matrix  $X$ , Target vector  $y$ , Threshold  $\theta$ )
2:   Compute the distance correlation matrix  $D$  for  $X$ 
3:   for each pair of features  $(a, b)$  in  $c$  do
4:     if  $c_{a,b} \geq \theta$  then
5:       Reduces redundancy
6:     end if
7:   end for
8:   Initialize the selected feature subset as  $NULL$ 
9:   while not all features  $f$  are evaluated do
10:    computed distance correlation for remaining features  $f$  with target  $y$ 
11:    for each feature  $f$  do
12:      if Distance correlation of  $f$  with  $y$  is significant then
13:        Add feature  $f$  to the selected feature subset
14:      end if
15:    end for
16:     $f \leftarrow$  Remaining features
17:  end while
18:  return the selected feature subset
19: end function

```

---

In the case of (AB, CD, E), 60 features are selected After applying the distance and correlation coefficients to nonlinear and linear features, while in (A, B), a total of 60 features form the Bonn dataset and the same number of features from the UCI-EEG dataset and CHB-MIT EEG dataset.

### E. Seizure detection module

In the seizure detection module, we explain binary and multi-class EEG epileptic seizure detection using a Stacking Ensemble Classifier (SEC) in a smart healthcare system.

1) *Proposed model*: After optimal feature selection, the proposed ESC is applied in the seizure detection module. The proposed model learns efficiently and differentiates the different patterns of the EEG. The proposed model is divided into two primary sections: base classifiers and meta-classifiers.

2) *Base classifiers*: The base classifiers consist of three models: random forest (RF), decision tree (DT), and extreme gradient-boosting (XGB). The RF model utilizes a collection of decision trees, each trained on a random data subset, to enhance predictive accuracy and control overfitting. It's particularly well-suited for handling complex datasets with multiple features [18]. The DT offers a single, comprehensive tree that segments the data into increasingly precise subsets using an intuitive, rule-based approach to categorization, while XGB advances the Gradient Boosting method by optimizing model complexity and regularization, significantly reducing the risk of overfitting [21]. The integration of different models aims to leverage their collective strengths, enhancing the robustness and reliability of the predictive framework in EEG ESD.

3) *Meta classifier*: The meta-classifier utilizes a logistic regression (LR) model. The diverse trees efficiently capture various combinations of the features. The regularization in LR helps to reduce the chances of overfitting. The output is binary, either 1 or 0, representing predictions from each tree. For multi-classification scenarios, we adapt the model to utilize multinomial logistic regression combined with the One-vs-All (OvA) strategy. In the OvA setup, for  $N = 3$ ,  $N$  binary classifiers are trained, each distinguishing a specific EEG epileptic seizure class from the rest. The parameters of our proposed model can be found in Table. V. Algorithm 2 highlights the model's operations. Moreover, ESC enhanced predictive accuracy by leveraging

**Algorithm 2** Stacking ensemble classifier

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```

1: Input: feature dataset  $D$ , Number of base models (RF, DT, XGB)  $N$ , Meta-classifier (LR)  $M$ 
2: Output: Stacked classifier  $E$ 
3: for each base model  $C_i$  in  $N$  do
4:   Train base model  $C_i$  on  $D_{train}$ 
5:   Get leaf indices  $L_i$  for  $C_i$  on  $D_{train}$ 
6: end for
7: Combine all leaf indices from all classifiers  $L = [L_1, L_2, \dots, L_N]$ 
8: One-hot encode leaf indices  $OHE(L)$ 
9: Train meta-model  $M$  on  $OHE(L)$ 
10: function STACKEDPREDICT( $x$ )
11:   for each base model  $C_i$  in  $N$  do
12:     Get leaf indices  $L_i^x$  for  $C_i$  on  $x$ 
13:   end for
14: Combine all leaf indices for input  $x$ ,  $L^x = [L_1^x, L_2^x, \dots, L_N^x]$ 
15: One-hot encode  $OHE(L^x)$ 
16: return PREDICT( $M, OHE(L^x)$ )
17: end function

```

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the strengths of multiple models (RF, DT, XGB) and combining their outputs; greater robustness against over-fitting as the ensemble captured diverse patterns in the EEG data which improved generalization, ensuring the model remains effective even if the characteristics of seizures vary among patients. Moreover, we integrate blockchain technology to ensure the integrity and

Table V: The hyper-parameters of the proposed model in EEG ES detection.

| Model  | Hyperparameter                                                                     |
|--------|------------------------------------------------------------------------------------|
| RF,XGB | max_features='sqrt', n_estimator=100                                               |
| DT     | max_depth=10,min_samples_leaf=10, bootstrap=True, n_jobs=-1, class_weight=balanced |
| MLR/LR | L1 and L2 penalties                                                                |

security of EEG data in the post-classification process. The EEG data is securely stored on the blockchain. The nodes of the blockchain authenticate the data and confirm its validity. The process assures the security of EEG data, allowing authorized access. The transparency and security offered by blockchain technology significantly enhance the privacy and security of the data in Smart healthcare systems for EEG ES detection

#### F. Explainable decision-making process

In the explainable decision-making process module, we discuss the explainability and interoperability of the proposed model decision-making process in a smart healthcare system for EEG epileptic seizure detection.

1) *Model interpretation:* After the classification of the proposed model, we perform the interoperability and explainability of the proposed model outcomes in a smart healthcare system (SHS) for EEG ES detection. It allows medical experts to validate the model's detection and make more informed and reliable decisions or recommendations regarding patient diagnosis and treatment. The SHAP interprets the model decision-making process and identifies the best feature contribution or feature importance. Additionally, it enables doctors and medical experts to gain deeper insights into the diagnostic patterns that the proposed model uses to identify seizures from the optimal feature data. The XAI used in recent work [7] in different ways such as;

1) Local explanation: The local explanation in XAI provides insight into the decision-making process of an AI model for a specific single instance or prediction.

2) Global explanation: The global explanation provides an overall understanding of the model's behavior across a broad set of instances. The two methods of local and global explanation are used in the proposed study for EEG epileptic seizure detection in the smart healthcare system.

## IV. EXPERIMENTAL RESULTS

This section discusses the experimental results, including comparative evaluations of epileptic seizure detection methods as well as interpretations and explanations of the proposed model.

### A. Ablation study

The performance evaluation is conducted on the three EEG datasets using the feature extraction methods described in subsections III-D. Fig.3 shows the validation accuracy of the proposed model applied different threshold values of the distance correlation (DC). The validation accuracy increases when the DC values increase from 0.2 to 0.6. However, after reaching 0.8, there was no significant change in the classification performance of the proposed model. At  $DC = 0.8$  in the Bonn EEG dataset (AB, CD, E), the mean accuracy reached 99.20%, while for the UCI dataset (A, B), it achieved 98.70%, and for the CHB-MIT EEG dataset (A, B), the accuracy is 99.60%. Similar performances are observed for the correlation coefficient (CC) vs validation accuracy of the proposed model. Moreover, Fig.2 shows the accuracy versus the number of estimators used by the base classifiers RF and XGB, it is observed from Fig.2 when the number of estimators increases from 10 to 100, the validation accuracy of the base classifiers increases. After  $n\_estimator = 100$ , there is no effect on the classification performance of the base classifiers.

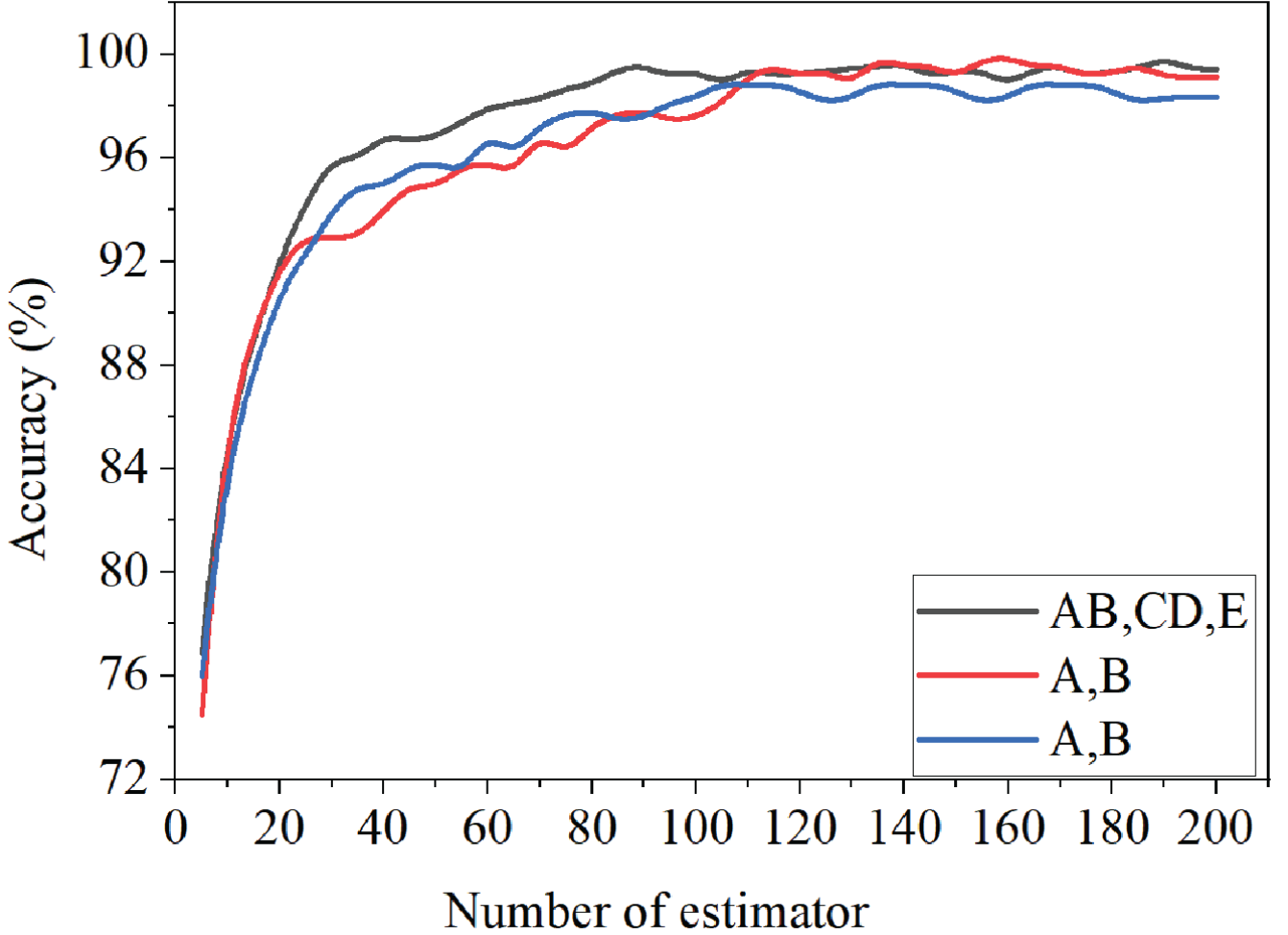


Figure 2: Accuracy vs  $n\_estimator$  of DS1, DS2, and DS3 case study: the validation accuracy increases when the number of estimators is equal to 100, while increasing the number of estimators has no effect on the performance of the base models.

Moreover, the classification performance of the Stacking Ensemble Classifiers (SEC) is evaluated and compared with other SECs, including SEC (RF)+LR, SEC(RF+XGB)+LR, and SEC (RF+XGB+DT)+LR, as shown in Fig.4. The evaluation is based on various performance metrics for EEG ES detection in a smart healthcare system, using the feature sets of the Bonn, UCI, and CHB-MIT EEG datasets (AB, CD, E), (A, B), and (A, B). The results show in Fig.4 that an increasing number of models also leads to improved performance. The proposed model outperforms other sibling ensemble classifiers by 2 to 3% based on validation accuracy (ACC), Recall (RE), specificity (Spec), and F1-score (F1). In contrast, the single-base ensemble classifiers exhibit the worst performance among all the models.

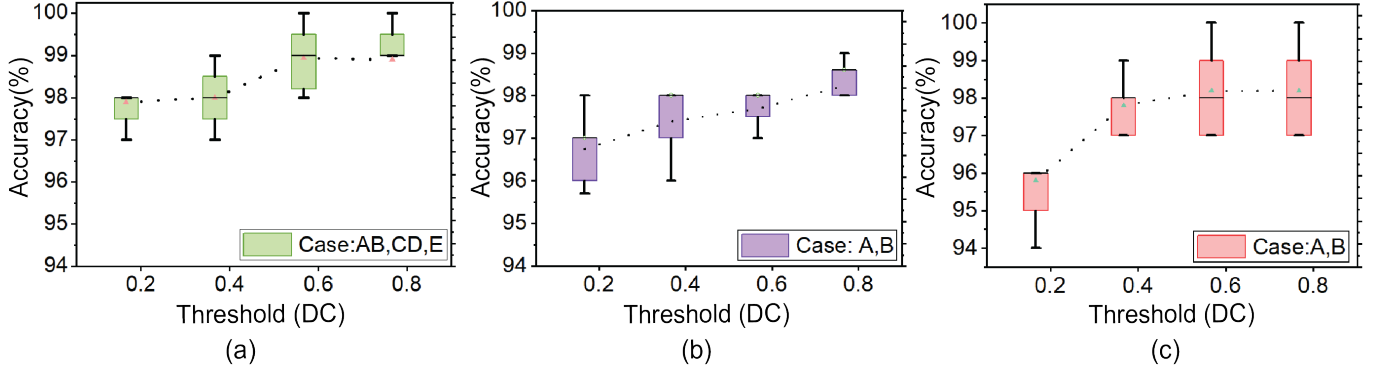


Figure 3: The ablation study of the distance correlation (DC) with validation accuracy, for the case study of (a) DS1, (b) DS2, and (c) DS3 in EEG epileptic seizure detection.

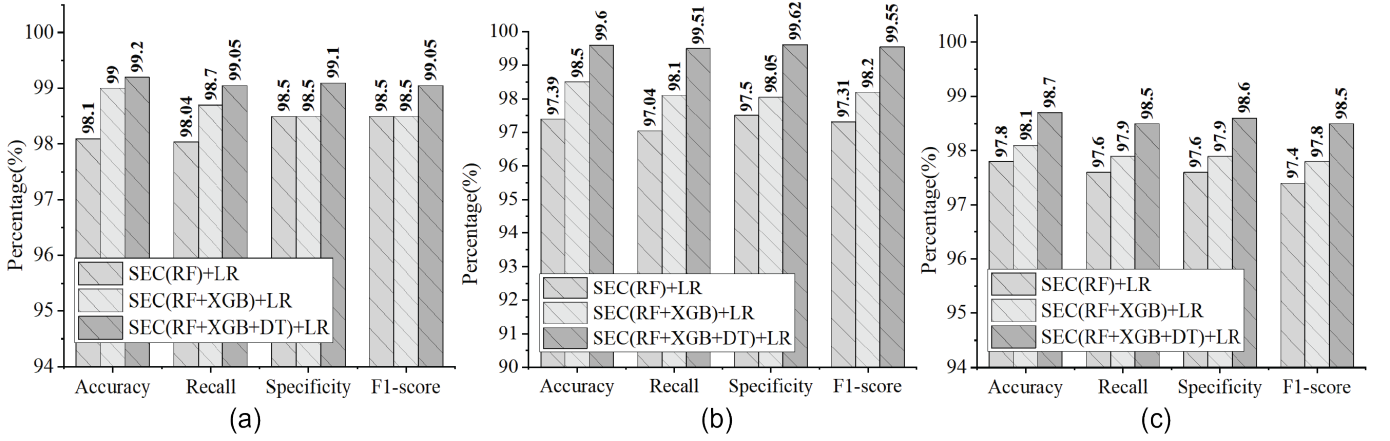


Figure 4: Comparative results of the proposed method with sibling classifier using (a) DS1, while (b) DS2 and (c) DS3.

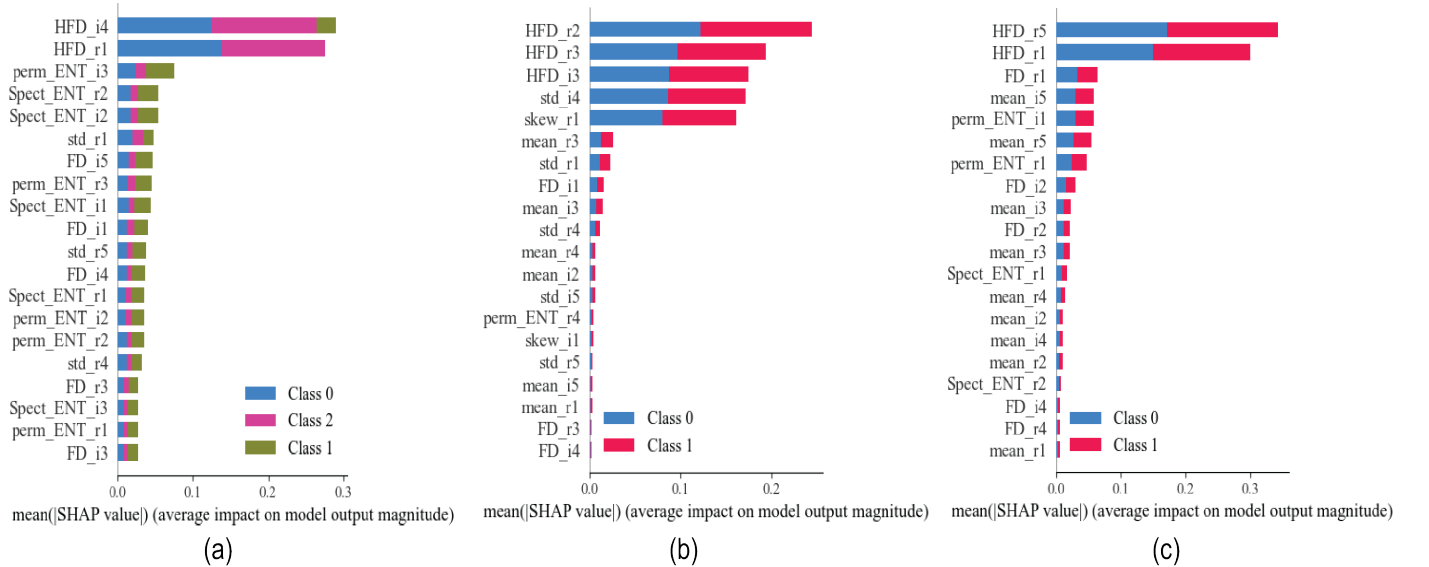


Figure 5: From left to right: decision plot of the proposed model using DS1, DS2 and DS3. Each SHAP decision plot illustrates how each feature contributes to shifting the model output from the base value (the average model output over the dataset) to the actual model output for EEG epileptic seizure detection.

### B. Comparison with other methods

The performance of the ensemble stacking classifier is also compared with single models such as DT, RF, XGB, and LR, as shown in Table VI. The proposed ensemble classifier achieved an accuracy of 99.20%, 99.60%, and 98.70% for the Bonn

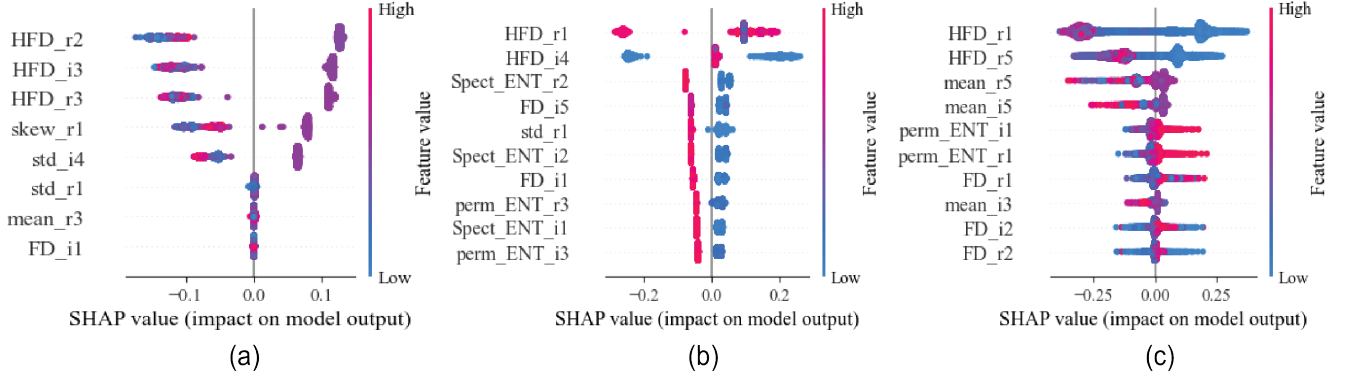


Figure 6: The impact of the features using the case study of the DS1, DS2, and DS3 of the proposed model using the SHAP summary plot, each point represents the SHAP analysis. The y-axis indicates the specific feature, while the x-axis shows the SHAP value. The color coding ranges from low to high, representing the feature's value in EEG epileptic seizure detection.

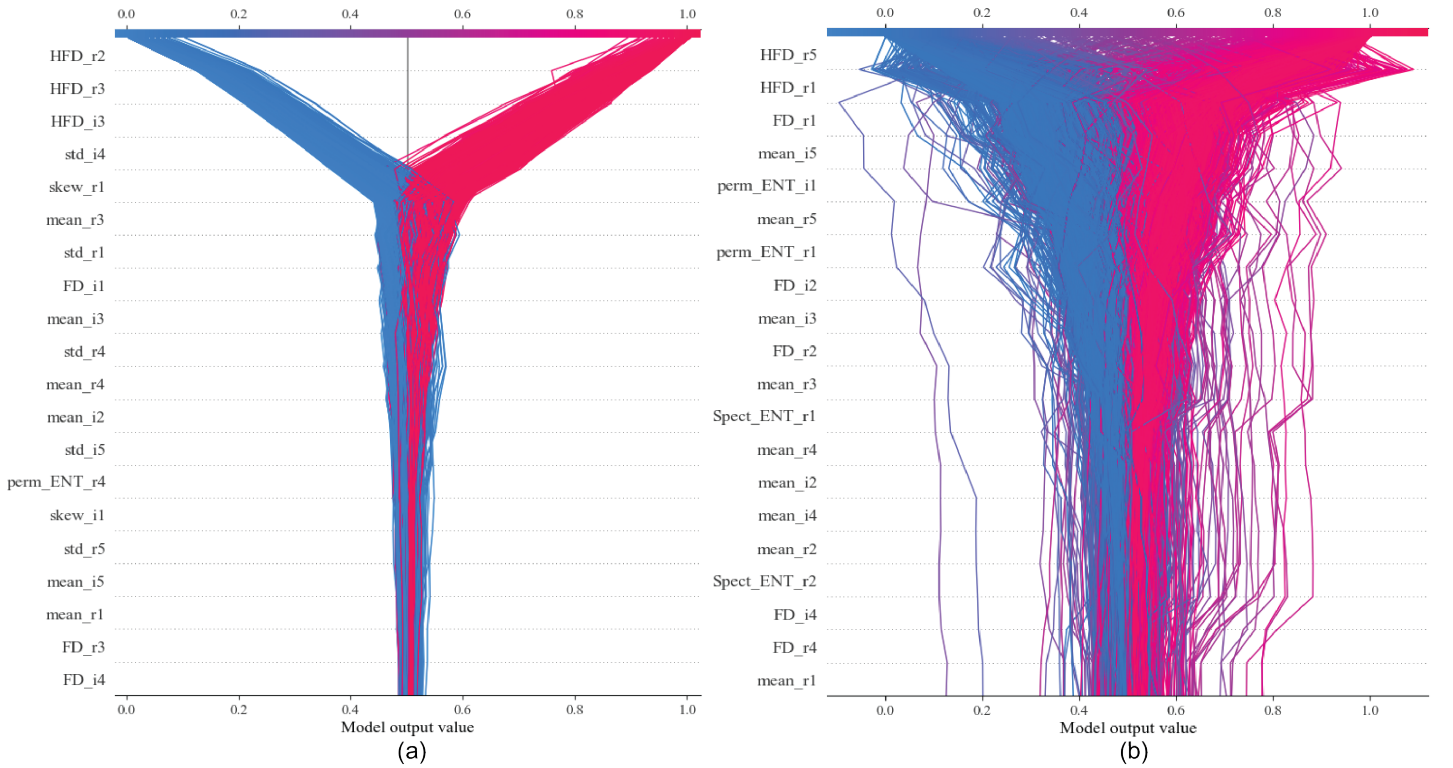


Figure 7: The force plot of the proposed model using (a) DS1, DS2 (b) DS3. The x-axis shows the features and the y-axis represents the model output value.

(AB, CD, E), CHB-MIT (A, B), and UCI (A, B) case studies in EEG epileptic seizure detection, while the Recall is 99.05%, 99.51%, and 98.50%, specificity is 99.10%, 99.62%, and 98.60%, and the F1-scores are 99.05%, 99.55%, and 98.50% for the respective case studies.

### C. Comparison with recent state-of-the-art methods

In this section, we discuss the comparative study of the proposed model with a recent machine-learning model (ML) using the same EEG dataset as shown in Table VII, IX, VIII. Table VII presents the Bonn EEG dataset using the same use case (AB, CD, E) and samples. The proposed model has the highest validation accuracy at 99.20%, while the proposed model in the recent literature [34] has the highest recall and specificity at nearly 100%. Moreover, Table VIII shows the comparative study of the CHB-MIT EEG data set. The stacking ensemble classifiers outperform other recent models, with validation accuracy of 99.60%, recall of 99.51%, and specificity of 99.62% using 10 patient EEG data sets. Finally, Table IX shows the comparative

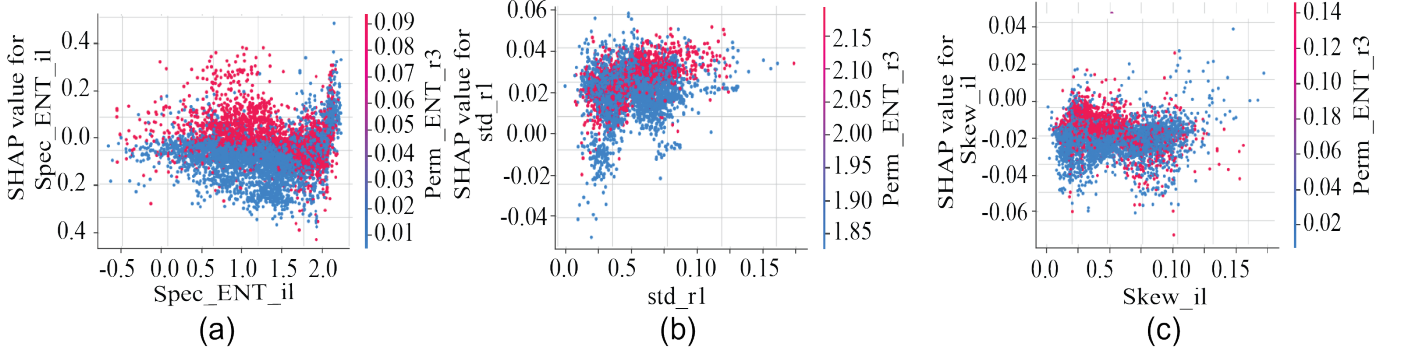


Figure 8: The dependency plot, (a) (b)(c) represents the different primary and secondary clinical features of the DS1, DS2, and DS3 experimental cases in smart health care systems in EEG Epileptic seizure detection.

Table VI: Comparison of single ML model with proposed ensemble classifiers using DS1, DS2, and DS3 for EEG epileptic seizure detection

|             | DS1 (A,B,C,D,E) |       |       |       |              | DS2 (A,B) |       |       |       |              | DS3 (A,B) |       |       |       |              |
|-------------|-----------------|-------|-------|-------|--------------|-----------|-------|-------|-------|--------------|-----------|-------|-------|-------|--------------|
|             | RF              | DT    | XGB   | LR    | SEC          | RF        | DT    | XGB   | LR    | SEC          | RF        | DT    | XGB   | LR    | SEC          |
| Accuracy    | 98.50           | 98.10 | 98.20 | 94.60 | <b>99.20</b> | 97.80     | 96.90 | 97.50 | 93.79 | <b>99.60</b> | 97.60     | 96.80 | 96.89 | 93.50 | <b>98.70</b> |
| Recall      | 98.41           | 98.05 | 98.06 | 94.10 | <b>99.05</b> | 97.30     | 98.20 | 96.30 | 93.45 | <b>99.51</b> | 97.68     | 96.67 | 96.73 | 92.90 | <b>98.50</b> |
| Specificity | 98.30           | 97.90 | 97.96 | 94.05 | <b>99.10</b> | 98.20     | 97.10 | 96.10 | 93.21 | <b>99.62</b> | 97.50     | 96.20 | 96.40 | 92.92 | <b>98.60</b> |
| F1-score    | 98.25           | 97.70 | 97.90 | 94.10 | <b>99.05</b> | 98.04     | 97.29 | 96.39 | 93.18 | <b>99.55</b> | 96.00     | 97.45 | 96.16 | 92.70 | <b>98.50</b> |

performance of the proposed model with a recent model using the UCI-EEG dataset. The proposed method obtained the best validation accuracy 98.70%, Recall 98.50%, and specificity 98.60%.

#### D. Explainability of the model using SHAP in smart healthcare system

In this section, the various plots of explainable artificial intelligence (XAI) are used to interpret the results generated by the proposed model utilizing different cases of the DS1, DS2, and DS3 with the SHAP method. The interpretation is conducted by various visual plots, including the SHAP decision plot (DP), summary plot (SP), force plot (FP), and dependent plot (DP). Fig.5 illustrates the feature contributions of the performance of each class. We observe that the HFD feature plays a significant role in the imaginary and real decomposition features. Conversely, the FD feature exhibits the lowest contribution to EEG epileptic seizure detection in smart healthcare systems. Additionally, in Fig.6, we present the DP for each case of DS1, DS2, and DS3. The color of the dots in these plots reflects the feature values, with pink dots indicating high feature values and blue dots representing lower values. The horizontal position of each dot on the x-axis signifies its impact on the model's output. The dots positioned on the right side indicate a positive impact, increasing the detection ratio of the seizure, while those on the left side signify a negative impact with a lower detection rate. The visualizations serve as valuable insights into the feature's importance when it comes to the model's ability to detect seizures, while fractal dimensions have minimal impact on the model's detection. Moreover, the SHAP summary plots of Fig.8 (a)(b)(c) indicate that features such as 'HFD\_r1' and 'perm\_ENT\_r1' exhibit a substantial impact on the performance of the model. The color gradient and width of the bands in the plots show the influence of various features with different instances in the feature data, indicating their importance in the model's decision-making process. Overall, all the SHAP plots are important for clinical decision-making, emphasizing the value of certain EEG signal characteristics in seizure detection.

## V. DISCUSSION

The current EEG epileptic seizure detection framework poses significant challenges including decomposing the EEG signals, where transformation coefficients have unpredictable changes with slight shifts in the EEG signal, often failing to represent the changes accurately and extracting meaningful features, selecting optimal features, reducing false positives, and lack of the explainability and interpretability of the decision-making process while maintaining the security and privacy of patient data.

To meet these challenges, the proposed study introduces a patient-independent novel framework consisting of an Explainable AI-based computer-aided ES detection system (XAI-CAESDs). The proposed framework has three key modules: feature engineering, detection, and an explainable decision-making process module in smart healthcare system. To address the privacy and security of biomedical EEG data, blockchain technology is used. Initially, for the noise removal and artifacts the Butterworth filter is applied. The Dual-Tree Complex Wavelet Transform (DTCWT) then decomposes the EEG signals, extracting both linear and nonlinear features (real and imaginary eigenvalue features). Optimal features are selected using correlation coefficients (CC) and distance correlation (DC), which are then input into the Ensemble Stacking Classifiers (ESC) for EEG ES detection.

Table VII: Comparison of state-of-the-art methods with proposed methods using DS1 for EEG epileptic seizure detection.

| References | Case    | No. of Samples | Methods                   | Accuracy (%) | Recall(%)  | Specificity (%) |
|------------|---------|----------------|---------------------------|--------------|------------|-----------------|
| [32]       | A,D,E   | 4097           | WT+Kmean clustering+MLPNN | 97.00        | 98.00      | 95.00           |
| [33]       | ABCD,E  | 4097           | TF+ANN                    | 98.10        | 99.05      | 95.10           |
| [34]       | ABCD,E  | 4097           | FUZZY APEn-SVM            | 99.15        | <b>100</b> | <b>100</b>      |
| [35]       | ABCD,E  | 4097           | DWT+LN+MLPNN              | 97.70        | 98.10      | 97.90           |
| This study | AB,CD,E | 4097           | DTCWT+LN+NL+CC+IDC+SEC    | <b>99.20</b> | 99.05      | 99.10           |

Table VIII: Comparison of state-of-the-art methods with proposed methods using DS2 for EEG epileptic seizure detection.

| References | Case | NO. of patients | Methods                     | Accuracy (%) | Recall(%)    | Specificity (%) |
|------------|------|-----------------|-----------------------------|--------------|--------------|-----------------|
| [36]       | A,B  | 8               | FFT+DWT+KNN                 | 99.10        | 99.05        | 99.01           |
| [37]       | A,B  | 23              | TF+WPD+FFT+TSK fuzzy system | 98.30        | 96.70        | 99.10           |
| [38]       | A,B  | 10              | RMS+R-square value +RF      | 94.00        | 96.00        | 96.00           |
| [39]       | A,B  | 23              | TF+DWT+SE+KNN               | 95.50        | -            | -               |
| This study | A,B  | 10              | DTCWT+LN+NL+CC+IDC+SEC      | <b>99.60</b> | <b>99.51</b> | <b>99.62</b>    |

In the XAI-enabled smart healthcare system (XAI-SHS), different SHAP methods are implemented to interpret the model's detection, enabling medical experts to make accurate and comprehensible decisions. Three EEG datasets (UCI, Bonn, and CHB-MIT) are utilized to have different case studies of EEG epileptic seizure detection.

Moreover, the results in Fig.3 show optimal performance at a distance correlation (DC) threshold value of 0.8. The DC value of 0.8 retains the most pertinent features for seizure detection while excluding the less informative ones, hence the models have high performance. Furthermore, it is observed from Fig.2, that the model performs best in accuracy and other performance metrics on using 100 estimators, indicating the ensemble classifiers have reached their learning capacity from the training data. Additionally, as the number of models in the ensemble increases, the performance also increases as shown in Fig.4. Because each model captures different patterns or features from EEG data. The integration of different models aims to leverage their collective strengths, enhancing the robustness and reliability of the predictive framework in EEG ESD. Table VI also indicates that the proposed model outperforms single-model approaches using the DS1, DS2, and DS3.

Comparative analyses with state-of-the-art methods demonstrate that the proposed stacking ensemble classifiers achieved the best validation accuracy, Recall, and specificity using different datasets, as shown in Table VII, IX, VIII.

Moreover, the explainability of the decision-making process of the proposed model indicates that the Higuchi Fractal Dimension (HFD) feature significantly contributes to EEG epileptic seizure detection, compared to other features, while the Fractal Dimension (FD) shows the lowest contribution due to the limitations in capturing the dynamic changes characteristic of epileptic EEG signals. Although the proposed framework offers significant advantages, but has some limitations. The proposed study utilized small datasets for the proposed case study. In the future, research will be conducted on more patients and large database in EEG epileptic seizure detection. Moreover, the framework incorporates blockchain technology as a theoretical enhancement for security and privacy, it has not been implemented in the current study.

## VI. CONCLUSION

In this study, we introduce a novel patient-independent framework for EEG epileptic seizure detection using EEG signals in smart healthcare system. The framework has three modules: feature engineering module, a seizure detection module, and an explainable decision-making process module in smart healthcare system. To ensure the privacy and security of biomedical EEG data, the blockchain is employed after detection. The EEG signals were pre-processed using Butterworth filtering and decomposed via Dual-Tree Complex Wavelet Transform (DTCWT), extracting both linear and non-linear features. The innovative framework integrates correlation coefficient and distance correlation feature selection methods. The proposed approach performed well on different patient data and outperformed other seizure detection methods and state-of-the-art methods, obtaining the best accuracy, Recall, and specificity. Moreover, in Explainable AI, the SHAP mechanism of XAI facilitated a clearer understanding of the model's decision-making processes for medical experts and generated understandable reports. The proposed patient-independent framework is highly suitable for clinical applications and ensures data security and

Table IX: Comparative analysis of the proposed methods using DS3 in EEG epileptic seizure detection .

| References | Case | No. of Samples | Methods                              | Accuracy (%)               | Recall(%)         | Specificity (%) |
|------------|------|----------------|--------------------------------------|----------------------------|-------------------|-----------------|
| [40]       | A,B  | 11500          | FT + 178 EEG features + KNN+ANN+SVM, | 97.01,96.10,95.40          | 96.00,96.10,94.20 | -               |
| [41]       | A,B  | 11500          | DCPA-EZ                              | 98.10                      | 97.80             | -               |
| [42]       | A,B  | 11500          | FFT + SOM-RBFnn                      | 97.40                      | -                 | -               |
| [43]       | A,B  | 11500          | 178 features + DT, RF, ANN, LDA      | 95.10, 96.40, 93.09, 94.10 | -                 | -               |
| This work  | A,B  | 11500          | DTCWT+LN+NL+CC+IDC+SEC               | <b>98.70</b>               | <b>98.50</b>      | <b>98.60</b>    |

privacy. In the future, the proposed framework will enhance clinical data with transfer learning and by implementing blockchain for EEG epileptic seizure detection.

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